

Physics-Informed Agentic Tuning of Gate-Defined Semiconductor Quantum Dots

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1 Introduction

Gate-defined quantum dots (QDs) are one of the most versatile and scalable platforms for engineering quantum electronic systems. By shaping electrostatic potentials with lithographically patterned metallic gates, experimentalists can confine individual electrons or holes, tune tunnel barriers, and assemble coupled arrays capable of coherent quantum operations. These devices have been realised across a wide range of material systems, from GaAs/AlGaAs two-dimensional electron gases (2DEGs) to Si/SiGe heterostructures, germanium nanowires, graphene nanoribbons, and transition-metal dichalcogenides, each offering distinct advantages in mobility, spin coherence, or fabrication compatibility (Lawrie et al. 2019, Tidjani et al. 2023, Banszerus et al. 2020)

Despite their conceptual simplicity, gate-defined QDs present a formidable control problem. Every gate electrode couples capacitively to multiple electrostatic degrees of freedom, and disorder from charge traps, interface states, and unintentional quantum dots introduces device-specific behaviour that cannot be predicted from design alone. As arrays grow from one or two dots to tens or hundreds, the dimensionality of the voltage space increases superlinearly, while the operational window, the region of gate space supporting stable charge configurations and controllable tunnel couplings, shrinks dramatically. Manual tuning, once feasible for single- and double-dot devices, becomes intractable for large arrays.

This section introduces the physical foundations of gate-defined quantum dots, reviews the state of the art in machine-learning-assisted tuning, and motivates the architectural challenges addressed by this work.

1.1 Gate-Defined Quantum Dots: Physical Principles and Control Landscape

Gate-defined QDs rely on electrostatic confinement: applying negative (electron) or positive (hole) voltages to patterned gates depletes carriers beneath them, forming quantum wells and tunnel barriers. The geometry, depth, and coupling of these potentials are controlled entirely by the applied voltages, enabling dynamic access to regimes ranging from the few-electron limit to many-body charge configurations.

Typical devices employ overlapping or multi-layer gate stacks to achieve fine-grained control of plunger potentials and barrier transparencies. The relevant energy scales include the charging energy $E_C \approx e^2/(2C)$, orbital level spacing $E_{\text{orb}} = \hbar\omega$, and tunnel coupling $t \propto \exp(-\alpha V_{\text{barrier}})$, all of which depend sensitively on gate voltages and local disorder. Crosstalk between gates, a consequence of the Maxwell capacitance matrix, means that adjusting one gate generally perturbs multiple dots and barriers simultaneously. This non-orthogonality is the central obstacle to scalable tuning.

As device complexity increases, so does the need for systematic, automated control. Even in modest double-dot systems, identifying the (1, 1) charge state requires navigating a non-convex, device-specific landscape where the operational region may occupy as little as 10^{-7} of the total

gate space (Schuff et al. 2026). For multi-dot arrays, the challenge is magnified: transition lines become dense and highly sloped, tunnel couplings vary nonlinearly, and disorder-induced features distort the expected Coulomb topology.

1.2 Machine Learning for Quantum Dot Tuning

Over the past decade, machine learning (ML) has emerged as a powerful tool for automating aspects of QD tuning. Early work focused on supervised classification of charge stability diagrams, using convolutional neural networks (CNNs) to distinguish single-dot, double-dot, barrier, and short-circuit regimes (Zwolak, 2018). Subsequent advances introduced ray-based classification (RBC), which uses one-dimensional line scans to infer state transitions with up to 70% fewer measurements than full 2D scans (Zwolak, 2021). These methods demonstrated that ML can replace human pattern recognition and accelerate coarse tuning.

More sophisticated approaches treat tuning as an optimisation problem. Gaussian-process (GP) models have been used to map the boundary hypersurface between conducting and non-conducting regions in high-dimensional gate spaces, enabling rapid identification of Coulomb peaks and double-dot regimes (Moon et al. 2020). Bayesian optimisation has been applied to tune tunnel barriers, while deep generative models such as conditional variational autoencoders (CVAEs) have enabled active learning strategies that prioritise informative measurements and reduce scan times by factors of 3–4 (Lennon et al. 2019).

The most striking demonstration to date is the fully autonomous tuning of a semiconductor spin qubit from a grounded device to Rabi oscillations (Schuff et al. 2026). This four-stage pipeline, combining GP hypersurface modelling, Bayesian optimisation, CNN-based Pauli spin blockade detection, and entropy-based readout-point selection, achieved median tuning times of 34 hours, outperforming human experts and revealing the primary bottleneck to be measurement integration time rather than algorithmic complexity.

1.3 The Automation Gap: Limitations of Existing Approaches

Despite these advances, current ML-based tuning systems exhibit several critical limitations that prevent their deployment in large-scale, heterogeneous quantum dot arrays:

- **Lack of explicit reasoning.** Most systems rely on black-box policies (e.g., reinforcement learning) that provide no interpretable rationale for their actions, making debugging and verification difficult for experimentalists.
- **Fixed measurement strategies.** Existing algorithms typically use fixed-resolution scans or heuristic thresholds, failing to reason about measurement cost or information gain.
- **Insufficient uncertainty quantification.** Many ML models output point estimates without calibrated confidence intervals, leading to brittle behaviour under distribution shift.
- **No online adaptation.** Models trained on one device often fail on another due to disorder-induced variability; few systems incorporate online learning or device-specific disorder inference.
- **No hierarchical task decomposition.** Tuning is inherently multi-stage, yet most algorithms treat it as a monolithic optimisation problem, leading to local minima and unstable behaviour.
- **Sim-to-real mismatch.** Synthetic datasets (e.g., QFlow lite, QDsim) accelerate development but do not fully capture the disorder landscape of real devices, creating a persistent reality gap.

These limitations collectively form the *automation gap*: the mismatch between what current ML systems can do and what is required for reliable, scalable, and interpretable quantum dot tuning.

1.4 Contributions of This Work

This blueprint proposes a clean-slate redesign of an autonomous tuning system that addresses the automation gap through a physics-first, uncertainty-aware, and hierarchically structured architecture. The system integrates:

- a hybrid POMDP planner with an LLM-based reasoning layer,
- an information-theoretic active sensing policy,
- a physics-aware perception stack combining CNNs, heuristics, and ensemble uncertainty,
- a Bayesian disorder-learning module that closes the sim-to-real gap,
- a strict hardware-agnostic device adapter layer, and
- a principled Human-in-the-Loop (HITL) safety gate.

Together, these components form a unified agentic framework capable of robust, interpretable, and device-adaptive tuning, with the long-term goal of enabling scalable control of large quantum dot arrays.